

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"Use the optical and SAR images together to identify built-up and water-covered regions."
Task Category:	OPTICAL_SAR
Specialist Tool:	Cross-Modal Optical-SAR Fusion Specialist (BigEarthNet / ISRO)
Reliability Score:	95.6% (HIGH TRUST)
Execution Time:	197.8 ms

1. Agent Synthesis & Visual-Language Answer

Cross-Modal Optical-SAR Multisensor Joint Information Extraction:

- Built-Up Regions: 10.1% scene coverage identified via SAR double-bounce microwave radar backscatter.
- Water Bodies: 15.8% scene coverage extracted via specular reflection low-backscatter signatures.
- Cloud Penetration Analysis: Note: Optical imagery exhibits 5.96% cloud obscuration. SAR microwave signals successfully penetrated cloud layer to reveal underlying ground geometry.
- Multisensor Synergy: Fusing optical spectral reflectance with SAR microwave structural geometry achieves 96.2% confidence feature delineation.

2. Auditable Agent Pipeline Execution Trace

- [Perception Layer]** Format & CRS validated. Mode: Co-registered Optical + SAR (2 image(s)). Status: PASSED
- [Query Engine]** Classified intent: 'OPTICAL_SAR'. Extracted entities: ['water', 'builtup']. Task: 'optical_sar'
- [Mission Planner]** Selected tool: 'Cross-Modal Optical-SAR Fusion Specialist' with params {'sar_double_bounce_thresh': 160, 'water_threshold': 40, 'cloud_thresh': 220}. Generated 7 node task-graph.
- [Specialist Tool Execution]** Executed specialist tool 'Cross-Modal Optical-SAR Fusion Specialist (BigEarthNet / ISRO)' with model confidence 96.2%.
- [Evidence Fusion]** Aligned 3 evidence streams. Cloud Conflict Resolved: Optical image had 5.96% cloud obscuration; SAR radar backscatter bypassed atmospheric clouds to confirm ground structures.
- [Trust & Uncertainty]** Calculated Reliability Score: 95.6% (HIGH TRUST). Cross-model agreement: 94.5%.
- [Discovery Engine]** Autonomous scan completed: Autonomous scan identified 2 'Beyond Your Query' discovery item(s).
- [Answer Generator]** Final evidence-linked response synthesized in 197.8 ms.